



HWA CHONG INSTITUTION
C1 Promotional Examination
Higher 2

NAME

CT GROUP

19S

CHEMISTRY

9729/01

Paper 1 Multiple Choice

2 October 2019

40 min

Additional Materials: Multiple Choice Answer Sheet
Data Booklet.

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Complete the information on the optical mark sheet (OMS) as shown below.

1. Enter your **NAME** (as in NRIC). _____

2. Enter the **PAPER NUMBER**. _____

3. Enter your **CT GROUP**. _____

4. Enter your **NRIC NUMBER** or
FIN Number

5. Now **SHADE** the corresponding
circles in the grid for
EACH DIGIT or **LETTER**



USE PENCIL ONLY FOR ALL ENTRIES ON THIS SHEET							
0	1	2	3	4	5	6	7
☐	☐	☐	☐	☐	☐	☐	☐

NRIC / FIN											
(S)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(A)	(K)	(U)	
(F)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(B)	(L)	(V)	
(G)	(2)	(2)	(2)	(2)	(2)	(2)	(2)	(C)	(M)	(W)	
(T)	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(D)	(N)	(X)	

There are **twenty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

- 1 Which species has more than two unpaired electrons in its outermost quantum shell?

A O^- **B** Zn **C** Co^{2+} **D** S

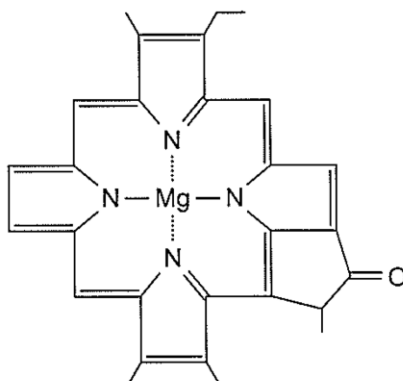
- 2 X, Y and Z are elements in Period 3 of the Periodic Table. The atomic radius of X is smaller than that of Y. The ionic radius of X is bigger than that of Y. The first ionisation energy of Z is higher than that of Y.

Which rows in the table are possible identities of elements X, Y, Z?

	X	Y	Z
1	P	Na	Ar
2	S	Al	Mg
3	Cl	Mg	S

A 1 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

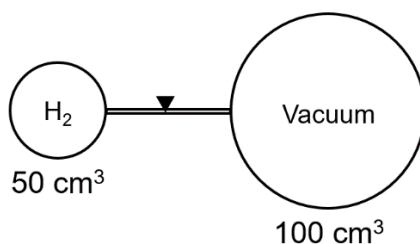
- 3 A simplified structure of a molecule of chlorophyll is as shown:



The magnesium atom is situated in the centre of a planar arrangement of nitrogen atoms. What does this structure suggest about the nature of the bonding around the magnesium atom?

- A** sp^3 hybridisation
B π bonding
C metallic bonding
D dative covalency

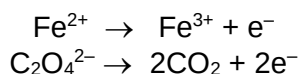
- 4 50 cm³ of H₂ gas at 220 °C and 2.05×10^4 Pa was connected to a 100 cm³ vessel which was initially a vacuum.



The valve was opened and the system was cooled to 18 °C.

What is the final pressure of the gas?

- A** 4.03×10^3 Pa
B 1.67×10^3 Pa
C 7.48×10^2 Pa
D 5.59×10^2 Pa
- 5 Consider the following half equations:



In an experiment, 20 cm³ of an acidified solution of 0.020 mol dm⁻³ of FeC₂O₄ is oxidised by 15 cm³ of 0.020 mol dm⁻³ KMnO₄.

What is the final oxidation state of manganese, given that both Fe²⁺ and C₂O₄²⁻ can be oxidised by KMnO₄?

- A** +1 **B** +2 **C** +3 **D** +4
- 6 When the salt (C₂H₅)₄N⁺Cl⁻ is treated with hydrogen chloride, a white ionic solid X which contains H and Cl in its anion is formed.

When X is dissolved in water, 20 cm³ of this solution requires 24 cm³ of 0.100 mol dm⁻³ NaOH for neutralisation.

Another 20 cm³ of the solution requires 48 cm³ of 0.100 mol dm⁻³ AgNO₃ for complete precipitation of chloride as AgCl.

Given that the above reactions are solely due to the anion in X, what is the formula of the anion in X?

- A** HCl₂⁻ **B** HCl₃²⁻ **C** HCl₄³⁻ **D** H₂Cl₃⁻

- 7 When 25 cm³ of 0.10 mol dm⁻³ KOH(aq) is added to 50 cm³ of 0.05 mol dm⁻³ of HNO₃(aq), the temperature of the mixture increases by 8.7 °C.

What is the temperature rise when 50 cm³ of 0.10 mol dm⁻³ KOH(aq) is added to 25 cm³ of 0.05 mol dm⁻³ of H₂SO₄(aq)?

- A** 2.2 °C **B** 4.4 °C **C** 8.7 °C **D** 17.4 °C

- 8 Which reaction is spontaneous at all temperatures?

	reaction	$\Delta H^\ominus / \text{kJ mol}^{-1}$
A	$(\text{NH}_4)_2\text{Cr}_2\text{O}_7(\text{s}) \rightarrow \text{Cr}_2\text{O}_3(\text{s}) + 4\text{H}_2\text{O}(\text{g}) + \text{N}_2(\text{g})$	-307.0
B	$\text{C}(\text{graphite}) + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g})$	-75.0
C	$\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow 3\text{H}_2(\text{g}) + \text{CO}(\text{g})$	+206.1
D	$\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$	+285.9

- 9 Which equation correctly represents the stated enthalpy change of reaction?

		enthalpy change of reaction
1	$\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$	$\Delta H_f^\ominus(\text{CO}_2)$
2	$\text{C}(\text{s}) + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g})$	$\Delta H_c^\ominus(\text{H}_2) + \Delta H_c^\ominus(\text{graphite}) - \Delta H_c^\ominus(\text{CH}_4)$
3	$\text{H}_2\text{S}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{S}(\text{s})$	$\Delta H_f^\ominus(\text{H}_2\text{O}) - \Delta H_f^\ominus(\text{H}_2\text{S})$

- A** 2 only **B** 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 10 Which statement best describes why increasing surface area of a solid reactant increases the rate of reaction?

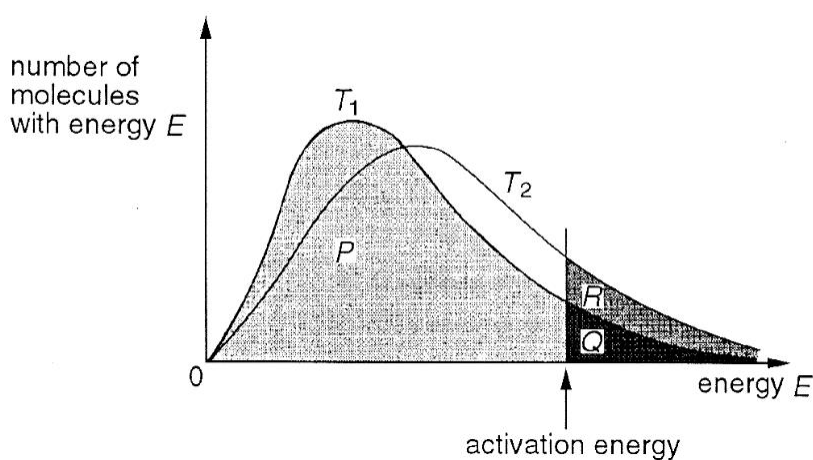
- A** The frequency of collision is increased.
B The rate constant of the reaction is increased.
C The activation energy of the reaction is reduced.
D More reactants have energy greater than the activation energy of the reaction.

11 Which gases are removed in the catalytic converter of vehicles?

- 1 CO
- 2 NO
- 3 SO₂

A 2 only B 1 and 2 only C 1 and 3 only D 2 and 3 only

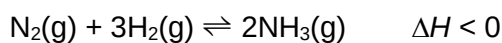
12 The distribution of the number of molecules with energy E is given in the sketch for two temperatures, T_1 and a higher temperature T_2 . The letters P , Q and R refer to the separate and differently shaded areas. The activation energy is marked on the energy axis.



Which expression gives the fraction of the molecules with energy greater than or equal to activation energy at T_2 ?

- A $\frac{Q}{P}$ B $\frac{Q+R}{P}$ C $\frac{Q+R}{P+Q}$ D $\frac{Q+R}{P+Q-R}$

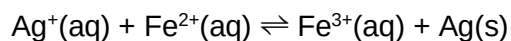
13 An equilibrium is represented by the following equation:



Which change would affect both the value of the equilibrium constant and the proportion of ammonia present at equilibrium?

- A adding a catalyst of finely divided iron
- B increasing the mass of hydrogen
- C increasing the pressure
- D reducing the temperature

14 The equation for the redox reaction between iron(II) and silver(I) is shown below:

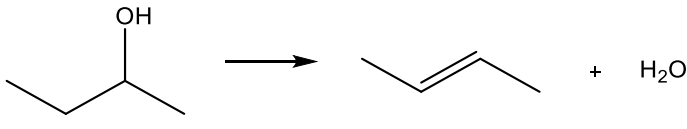
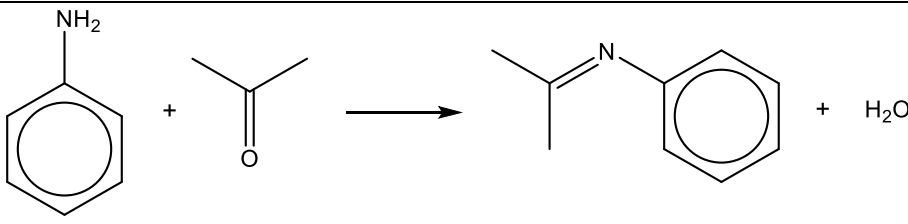
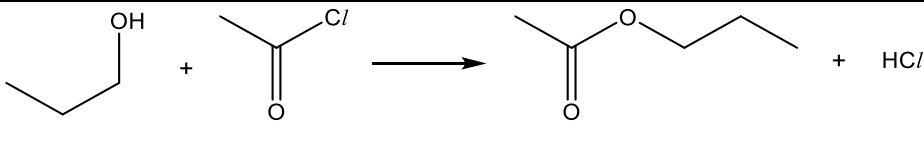
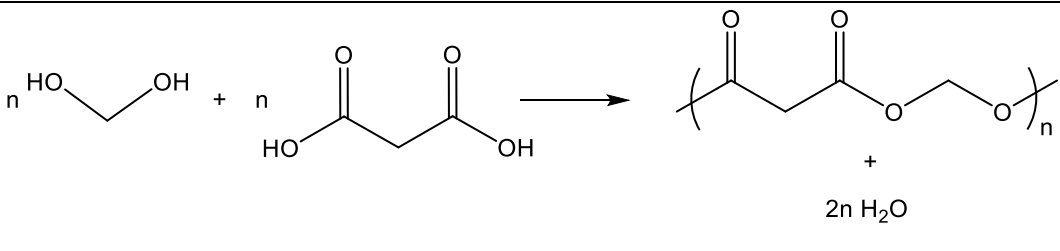


The equilibrium constants were measured to be 3.1 at 25 °C and 1.2 at 35 °C. Which statements are correct?

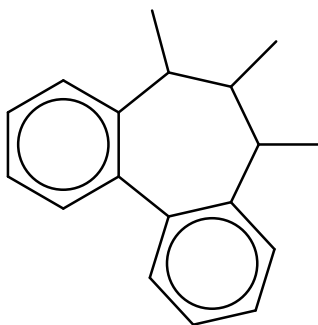
- 1 The addition of sodium chloride solid has no effect on the position of the equilibrium.
- 2 The equilibrium constant K_c has no units.
- 3 The forward reaction is exothermic.

A 1 only B 3 only C 1 and 3 only D 2 and 3 only

15 Which is **not** a condensation reaction?

A	
B	
C	
D	

16 How many chiral centres are present in the structure below?



A 0

B 2

C 3

D 6

17 Which could be possible product(s) of the reaction between methane and chlorine in the presence of sunlight?

A CH_3Cl

B CHCl_2CH_3

C $\text{CH}_2\text{ClCH}_2\text{Cl}$

D All of the above

18 Ethene is reacted with bromine water containing chloride ions. Which products are formed?

1 $\text{Cl}(\text{CH}_2)_2\text{Cl}$

2 $\text{Br}(\text{CH}_2)_2\text{Cl}$

3 $\text{Br}(\text{CH}_2)_2\text{OH}$

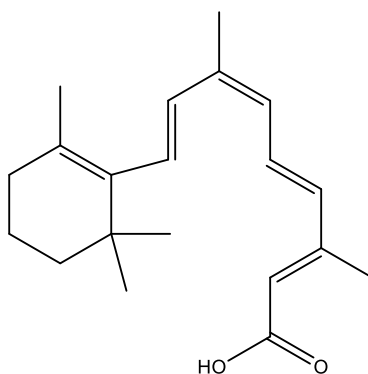
A 3 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3

- 19 *Alitreinoin* is an antineoplastic agent used to treat lesions caused by the AIDS-related disease, Kaposi's sarcoma. Its structure is shown below.

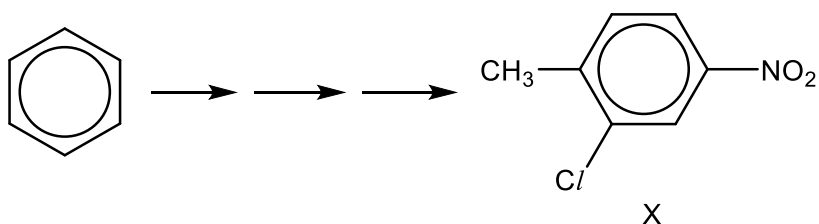


Alitreinoin

Which statement is **incorrect**?

- A *Alitreinoin* may exhibit sixteen different stereoisomers.
- B In the presence of excess bromine and uv light, *Alitreinoin* will only undergo free radical substitution.
- C The product of the reaction of *Alitreinoin* with cold dilute potassium manganate(VII) solution is chiral.
- D Addition of hot acidic potassium manganate(VII) solution to *Alitreinoin* will result in the effervescence of carbon dioxide gas.

- 20 Benzene can be converted to X via a three-step synthetic route.



Which sequence of steps leads to the highest yield of X?

	step 1	step 2	step 3
A	conc. HNO_3 , conc. H_2SO_4 heat	Cl_2 , AlCl_3 , warm	CH_3Cl , FeCl_3 , warm
B	conc. HNO_3 , conc. H_2SO_4 heat	CH_3Cl , FeCl_3 , warm	Cl_2 , AlCl_3 , warm
C	CH_3Cl , FeCl_3 , warm	Cl_2 , AlCl_3 , warm	conc. HNO_3 , conc. H_2SO_4 heat
D	CH_3Cl , FeCl_3 , warm	conc. HNO_3 , conc. H_2SO_4 heat	Cl_2 , AlCl_3 , warm

~END OF PAPER~